Software Requirements Engineering

Handout # 2

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Kinds of Software Requirements

- Functional requirements
- Non-functional requirements
- Domain requirements
- Inverse requirements
- Design and implementation constraints

Non-Functional Requirements

Non-Functional Requirements - 1

- Most non-functional requirements relate to the system as a whole.
- They include constraints on timing, performance, reliability, security, maintainability, accuracy, the development process, standards, etc.

Non-Functional Requirements - 2

- They are often more critical than individual functional requirements
- Capture the emergent behavior of the system, that is they relate to system as a whole

Non-Functional Requirements - 3

- Must be built into the framework of the software product
- Failure to meet a non-functional system requirement may make the whole system unusable

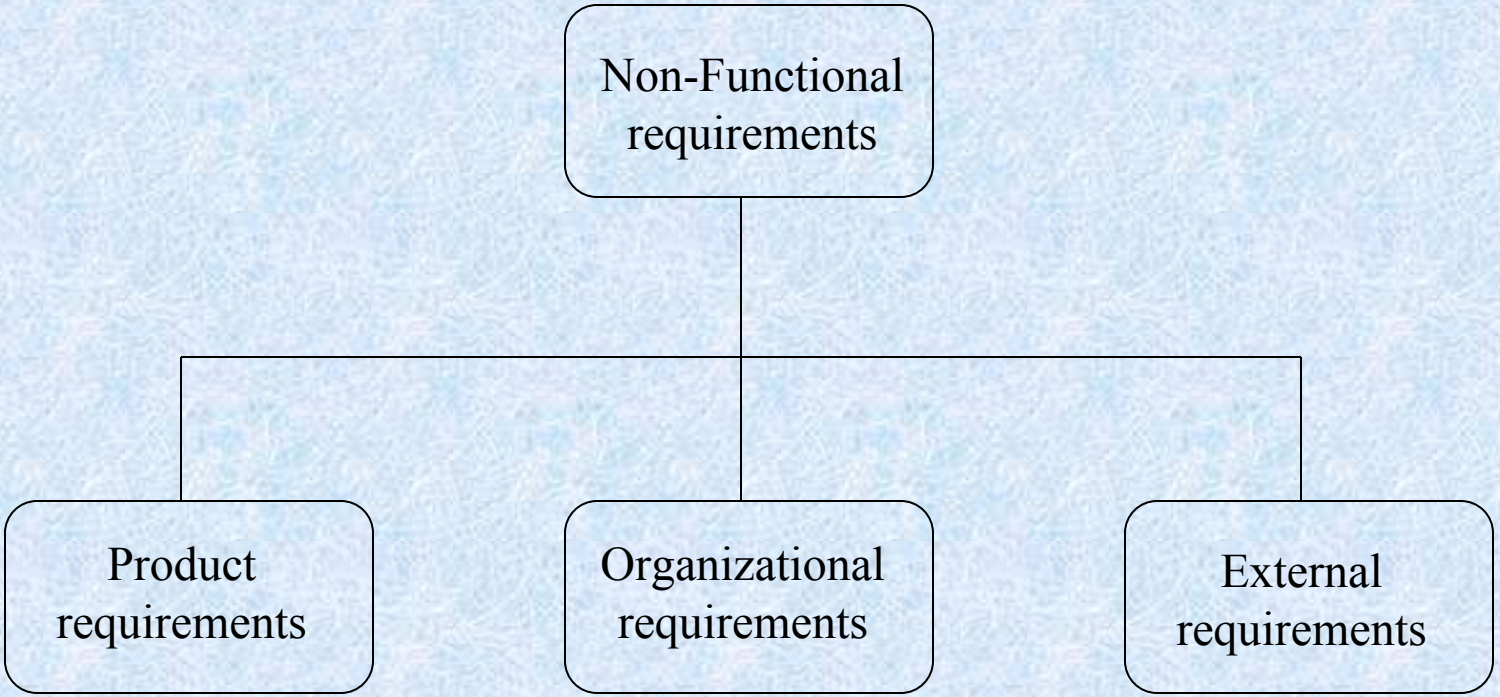
Non-Functional Requirements - 4

- For example, if an aircraft system does not meet reliability requirements, it will not be certified as ‘safe’
- If a real-time control system fails to meet its performance requirements, the control functions will not operate correctly

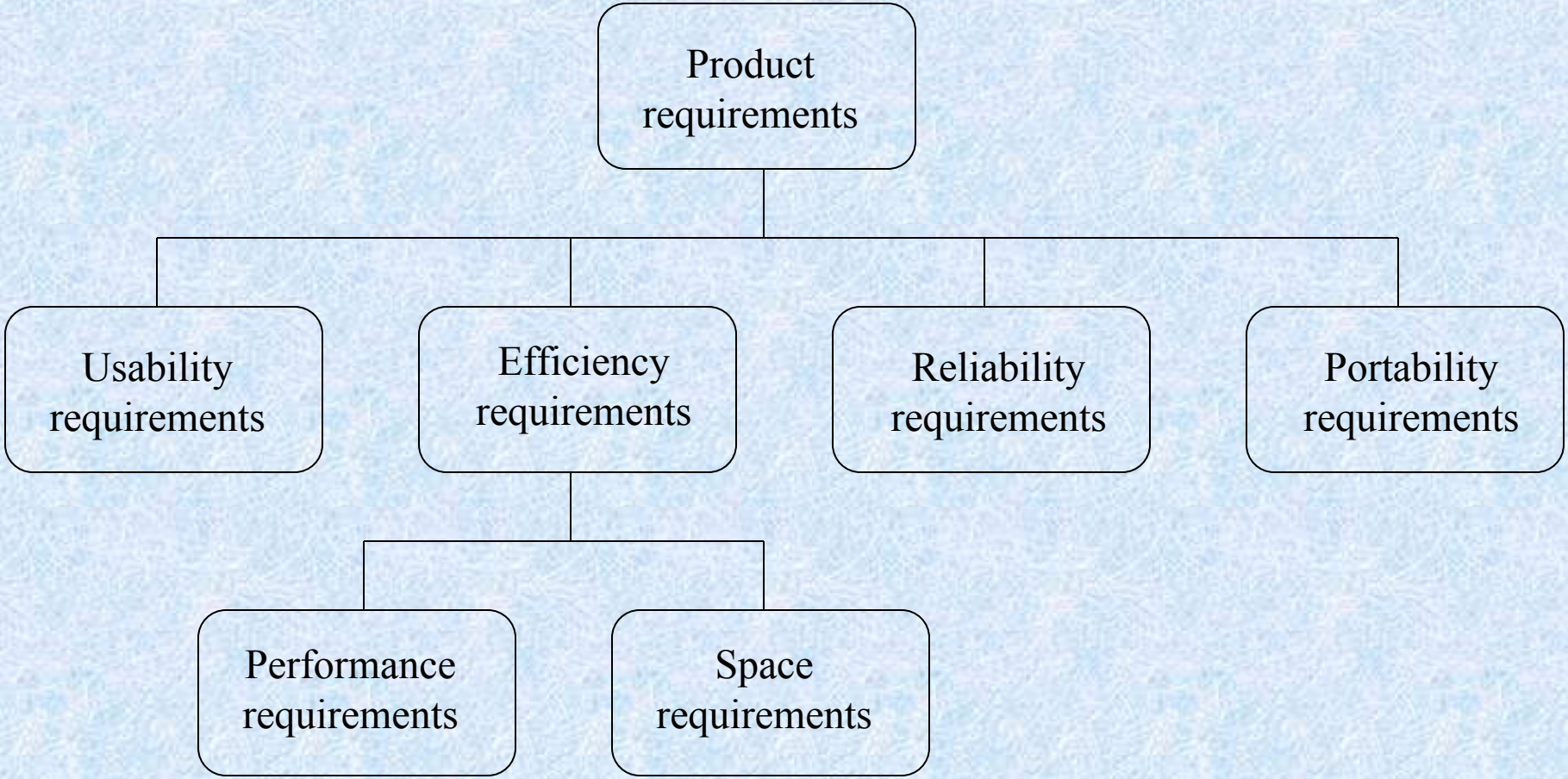
Non-Functional Requirements - 5

- Non-functional requirements arise through user needs,
  - because of budget constraints,
  - because of organizational policies,
  - because of the need of interoperability with other software and hardware systems, or
  - because of external factors such as safety regulations, privacy legislation, etc.

Non-Functional Requirements



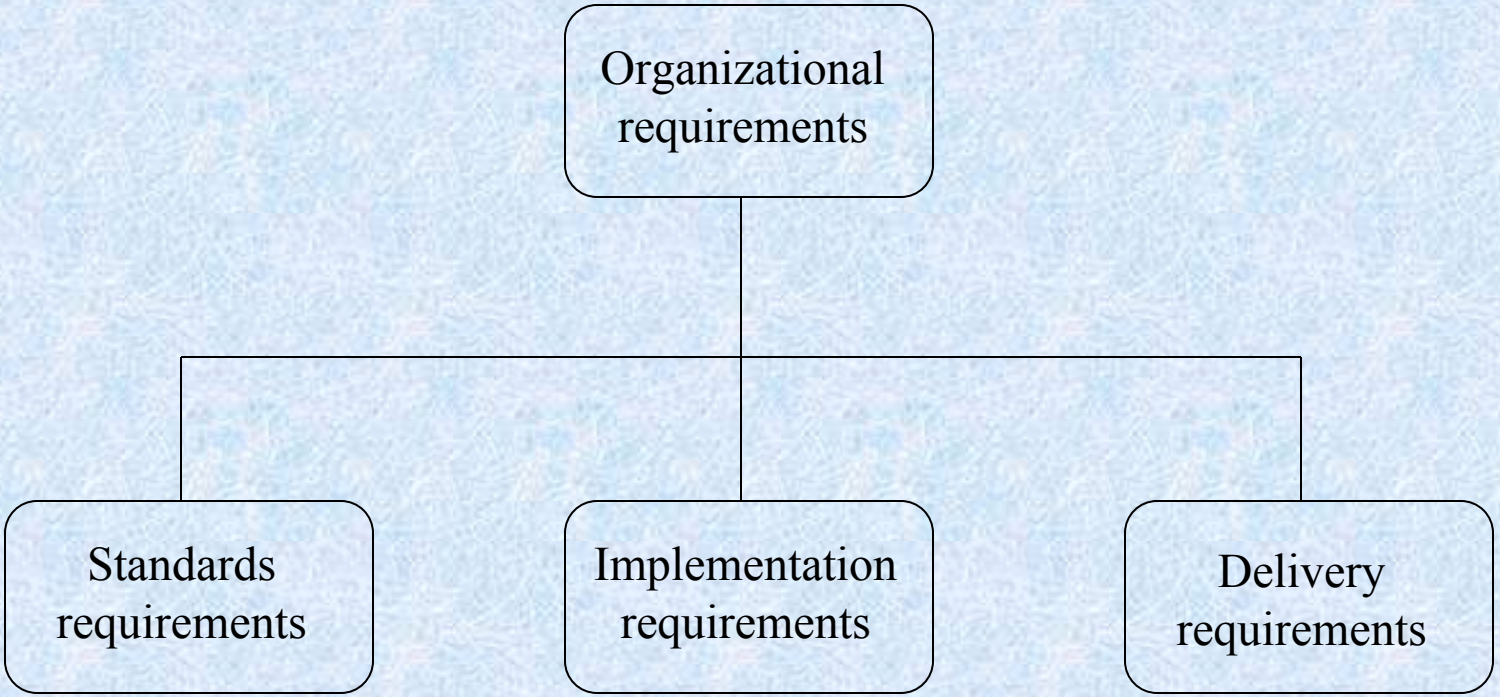
Product Requirements



Product Requirements Examples

- The system shall allow one hundred thousand hits per minute on the website
- The system shall not have down time of more than one second for continuous execution of one thousand hours

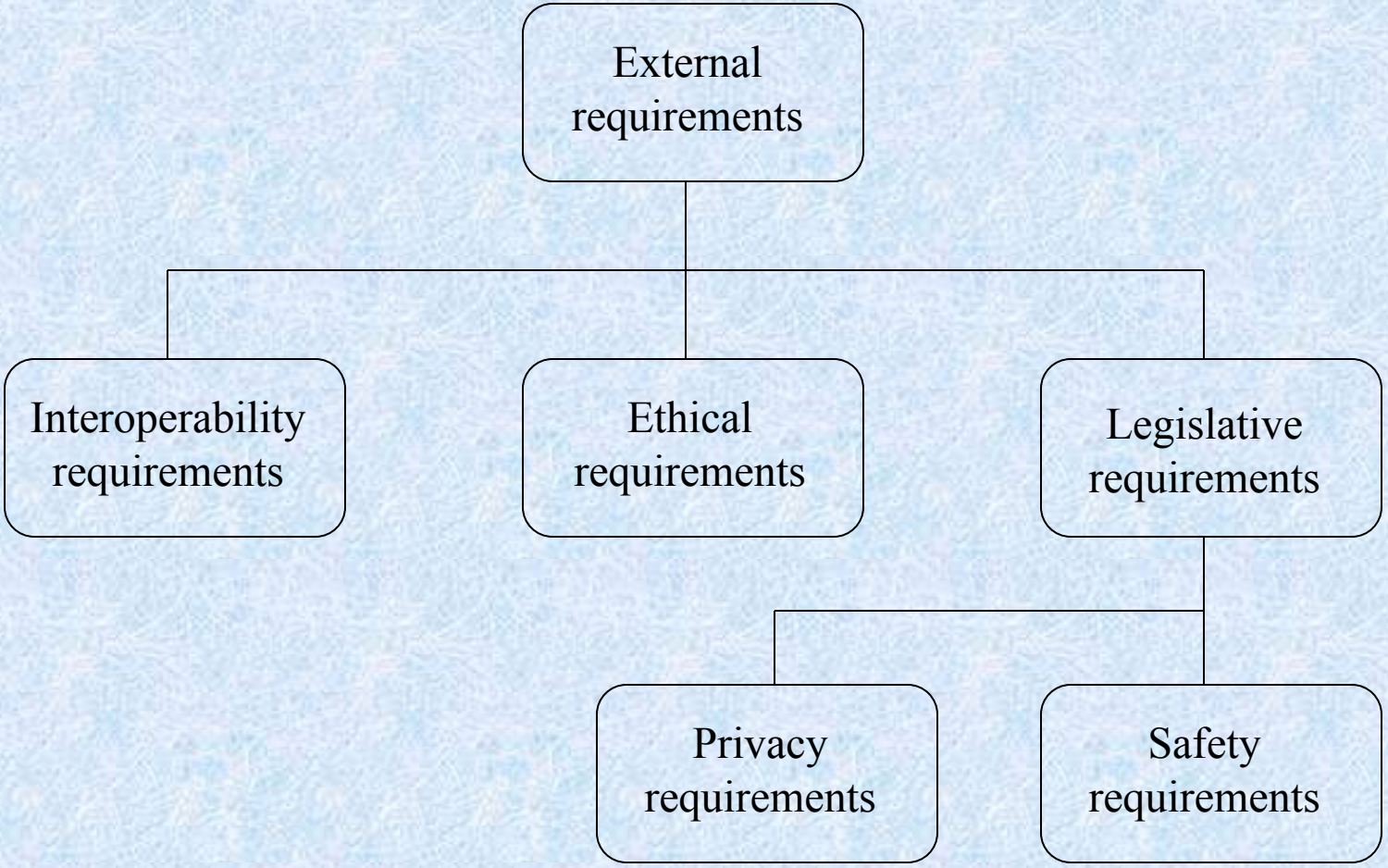
Organizational Requirements



Organizational Requirements Examples

- The system development process and deliverable documents shall conform to the MIL-STD-2167A
- Any development work sub-contracted by the development organization shall be carried out in accordance with Capability Maturity Model

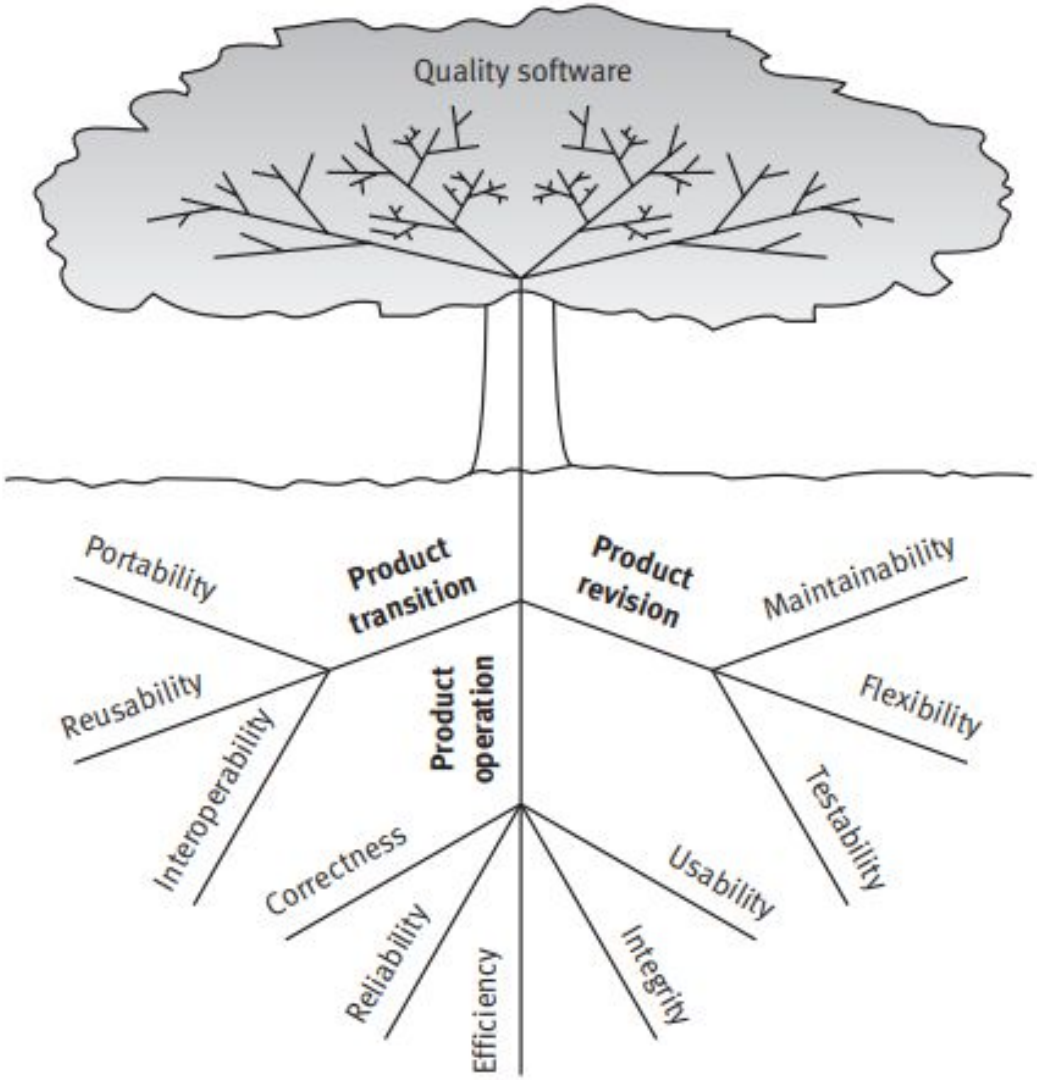
External Requirements



External Requirements Examples

- The system shall not disclose any personal information about members of the library system to other members except system administrators
- The system shall comply with the local and national laws regarding the use of software tools

Macall's Quality Factors Model



Observations on Non-Functional Requirements - 1

- Non-functional requirements can be written to reflect general goals for the system. Examples include:
  - **Ease of use**
  - **Recovery from failure**
  - **Rapid user response**

Observations on Non-Functional Requirements - 2

- Goals are open to misinterpretation
- Objective verification is difficult
- Distinction between functional and non-functional is not always very clear.
  e.g. Modified data in a database should be updated for all users accessing it within 2 seconds.

Observations on Non-Functional Requirements - 3

- Non-functional requirements should be written in a quantitative manner as much as possible, which is not always easy for customers
- For some goals, there are no quantitative measures, e.g., maintainability

Observations on Non-Functional Requirements - 4

- Goals can be useful to designers and developers, as they give clues to them about priorities of the customers

Observations on Non-Functional Requirements - 5

- Chances of conflicts within non-functional requirements are fairly high, because information is coming from different stakeholders.
- For example, different stakeholders can give different response times or failure tolerance levels, etc.

Observations on Non-Functional Requirements - 6

- Some negotiations must be done among different stakeholders, to achieve an agreement in these situations

Observations on Non-Functional Requirements - 7

- Non-functional requirements should be highlighted in the requirements document, so that they can be used to build the architecture of the software product

Summary

- Discussed different aspects of the non-functional requirements
- Non-functional requirements capture very important emergent behavior of the automated system
- Due importance, time, and resources should be given to non-functional requirements

References

- 'Requirements Engineering: Processes and Techniques' by G. Kotonya and I. Sommerville, John Wiley & Sons, 1998
- Software Requirements: Objects, Functions, and States by A. Davis, PH, 1993
- Software Engineering 6th Edition, by I. Sommerville, 2000
- Software Engineering 5th Edition, by R. Pressman

Assignment 1

- The Macall's Model is considered as a benchmark in industry for non-functional requirements. You are required to do your research of the model and explain it in your own words.
- Identify at least two practical examples of each non-functional requirement discussed in this lecture
- *Submission: Second class of next Week.*